

OPTIMISATION PROBLEMS: A GEOMETRIC APPROACH

Mauricio Labadie, PhD

Seminario de Finanzas Cuantitativas con Python

Facultad de Ciencias - UNAM

Mexico City, 30 October 2020

Exact solution for portfolio optimisation

- Let $x = \begin{bmatrix} S_1 \\ \vdots \\ S_N \end{bmatrix}$ and $\beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_N \end{bmatrix}$
- We have already found an optimal solution for a portfolio hedge that is beta and delta neutral:
 - If $N = 2$ then the problem is to find $x = \begin{bmatrix} S_1 \\ S_2 \end{bmatrix}$ such that

$$\begin{bmatrix} 1 & 1 \\ \beta_1 & \beta_2 \end{bmatrix} \begin{bmatrix} S_1 \\ S_2 \end{bmatrix} = - \begin{bmatrix} S_0 \\ \beta_0 S_0 \end{bmatrix}$$

- If $\beta_1 \neq \beta_2$ then the solution exists and is unique:

$$\begin{bmatrix} S_1 \\ S_2 \end{bmatrix} = - \begin{bmatrix} 1 & 1 \\ \beta_1 & \beta_2 \end{bmatrix}^{-1} \begin{bmatrix} S_0 \\ \beta_0 S_0 \end{bmatrix}$$

Numerical solution for portfolio optimisation

- Let $x = \begin{bmatrix} S_1 \\ \vdots \\ S_N \end{bmatrix}$, $\beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_N \end{bmatrix}$ and $\mathbb{I} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$

- Define

$$f(x) := f_{\text{delta}}(x) + f_{\text{beta}}(x)$$

- where

$$\begin{aligned} f_{\text{delta}}(x) &:= (\mathbb{I}^T x + S_0)^2 \\ f_{\text{beta}}(x) &:= (\beta^T x + \beta_0 S_0)^2 \end{aligned}$$

- The cost function $f(x)$ is a quadratic function in x i.e. its gradient is linear in x and its Hessian matrix (second derivatives) is constant
- If $N \geq 2$ then $f(x)$ has at least one global minimum

$$f(x_{\min}) = 0$$

- where

$$\begin{aligned} \mathbb{I}^T x_{\min} + S_0 &= 0 \\ \beta^T x_{\min} + \beta_0 S_0 &= 0 \end{aligned}$$

Numerical solution for portfolio optimisation with constraints

- If $N > 2$ then there are infinitely many solutions to the problem

$$f(x) = 0$$

- We are interested in the solution with “smallest weights”, that is x with the smallest norm $\|x\|$
- We could try to minimise norm under the constraints of beta and delta neutrality:

➤ Minimise

$$g(x) = \|x\|^2$$

➤ Under the constraints

$$\begin{aligned} h_1(x) &= \mathbb{I}^T x + S_0 = 0 \\ h_2(x) &= \beta^T x + \beta_0 S_0 = 0 \end{aligned}$$

- This is a standard problem that can be solved using Lagrange Multipliers
- But it does not solve the issue in we found in practice of “very large weights” when $N = 2$
 - The hedge of +10 mnUSD BBVA.MC was +248 mnUSD CAC40 and −258 mnUSD DAX
 - We need to find a solution that the portfolio manager likes: one-sided hedge

Numerical solution for portfolio optimisation with regularisation

- Let $x = \begin{bmatrix} S_1 \\ \vdots \\ S_N \end{bmatrix}$, $\beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_N \end{bmatrix}$ and $\mathbb{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$

- Define

$$f(x; \varepsilon) := f_{\text{delta}}(x) + f_{\text{beta}}(x) + f_{\text{penalty}}(x; \varepsilon)$$

- where

$$\begin{aligned} f_{\text{delta}}(x) &:= (\mathbb{1}^T x + S_0)^2 \\ f_{\text{beta}}(x) &:= (\beta^T x + \beta_0 S_0)^2 \\ f_{\text{penalty}}(x; \varepsilon) &= \varepsilon \|x\|^2 \end{aligned}$$

- The **parameter** $\varepsilon \geq 0$ will allow the portfolio manager to **control** the degree of “one-sidedness” of the optimal solution
 - No constraints → we solve the problem without the need of Lagrange Multipliers
 - The solution cannot be perfectly beta-neutral and delta-neutral, as before
 - But the portfolio manager will like this solution more than the exact solution

Minimum-variance portfolio as an optimisation problem with constraints

- Let Q be a variance-covariance matrix i.e. symmetric and positive-definite
 - At this stage, just think about a matrix such that $x^T Q x > 0$ for all $x \in \mathbb{R}^N - \{0\}$
 - The **minimum-variance portfolio** is a vector of weights or notionals $x \in \mathbb{R}^N$ that minimises the variance, defined as
- $$f(x) = x^T Q x$$
- Of course, the global minimum is $f(x) = 0$, which can only occur when $x = 0$
 - But this is a boring solution: we want a portfolio that is not zero, hence we need to add some restrictions:
 - Non-negative weights: $S_n \geq 0$ for all $n = 1, \dots, N$
 - Unitary portfolio in L_2 -norm or L_1 -norm, respectively:

$$\|x\|_2 = \left(\sum S_n^2 \right)^{1/2} = 1$$

$$\|x\|_1 = \sum |S_n| = 1$$

Markowitz portfolio as an optimisation problem with constraints

- The **Markowitz portfolio** is the minimum-variance problem before with an extra restriction on the returns of the portfolio

- Let $r = \begin{bmatrix} r_1 \\ \vdots \\ r_N \end{bmatrix}$ be the vector of returns, and define a target return r_{target}

- The problem is now to minimise the variance

$$f(x) = x^T Q x$$

under the following restrictions:

- Non-negative weights
- Unitary portfolio in L_2 -norm or L_1 -norm
- **New restriction:** portfolio return equals to the given target return

$$r^T x = r_{target}$$

- Both problems, minimum variance portfolio and Markowitz portfolio, are standard minimisation problems with restrictions
 - As such, they can be solved using **Lagrange multipliers**

Lagrange Multipliers

Statement of the problem

- Suppose we have the following minimisation problem:
 - Minimise the (scalar-valued) function $f(x): \mathbb{R}^N \rightarrow \mathbb{R}$
 - Under the constraint $g(x) = 0$
- If p is the minimum of $f(x)$ in the $N - 1$ dimensional surface $S = \{x: g(x) = 0\}$ then there exists a real number λ such that

$$\nabla f(p) + \lambda \nabla g(p) = 0$$

- Define the Lagrangian

$$L(x, \lambda) := f(x) + \lambda g(x)$$

- Solving the constrained problem is equivalent to solving the unconstrained problem:
 - Minimise $L(x, \lambda): \mathbb{R}^{N+1} \rightarrow \mathbb{R}$
- This is true because:

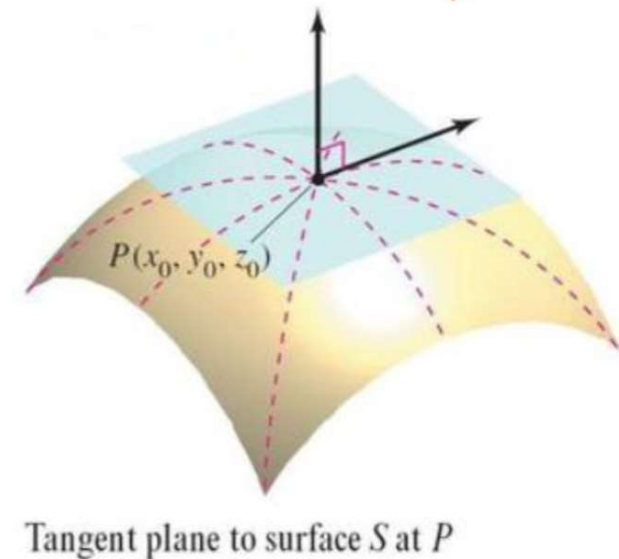
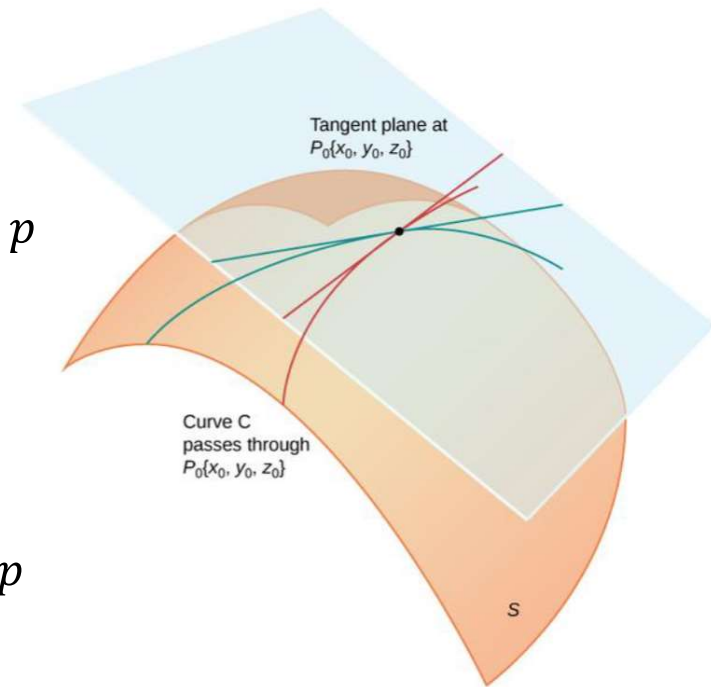
$$\begin{aligned}\frac{\partial L}{\partial x}(x, \lambda) &= \nabla f(x) + \lambda \nabla g(x) \\ \frac{\partial L}{\partial \lambda}(x, \lambda) &= g(x)\end{aligned}$$

Lagrange Multipliers Proof

- Let $p \in S$ and $t \rightarrow c(t)$ be any curve on S with $c(0) = p$
- Define $u: R \rightarrow R$ as $u(t) = f(c(t))$
- If $f|_S$ has a local minimum at $p \in S$ then $u'(0) = 0$
- By the Chain Rule we have $u'(0) = \nabla f(p)^T c'(0)$
- Hence $\nabla f(p)$ is orthogonal to all tangent vectors of S at p

$$\nabla f(p) \perp T_p S$$
- Define $v: R \rightarrow R$ as $v(t) = g(c(t))$
- Since $c(t) \in S$ then $v(t) \equiv 0$, which implies $v'(0) = 0$
- By the Chain Rule again $v'(0) = \nabla g(p)^T c'(0)$
- In consequence $\nabla g(p) \perp T_p S$
- But $T_p S$ is an $N - 1$ dimensional plane
- Therefore $\nabla f(p)$ and $\nabla g(p)$ must be collinear:

$$\nabla f(p) + \lambda \nabla g(p) = 0$$

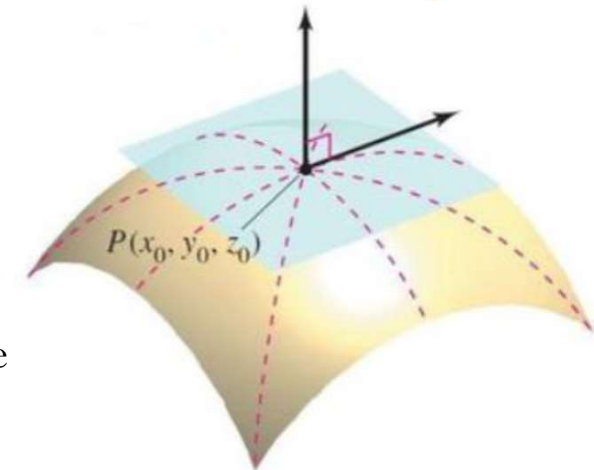
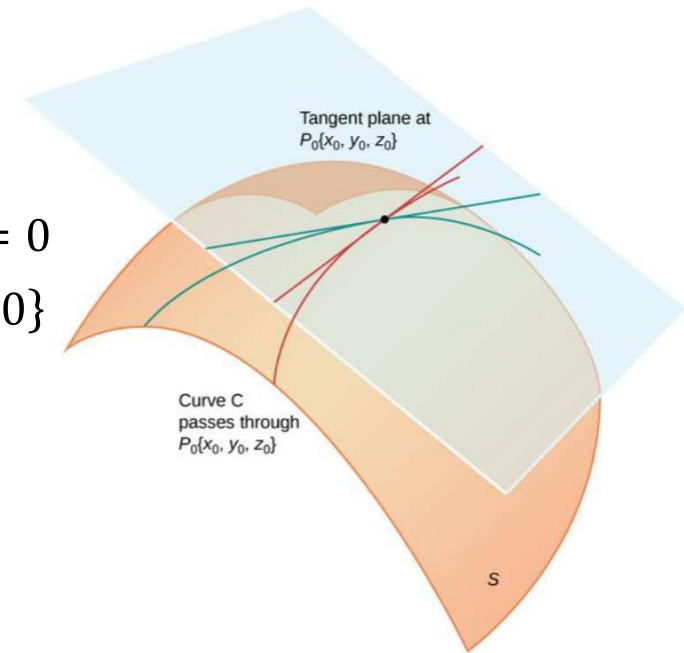


Lagrange Multipliers

Several constraints

- For any $k = 1, \dots, K$ assume we have a constraint $g_k(x) = 0$
- Define the $N - 1$ dimensional surface $S_k = \{x: g_k(x) = 0\}$
 - Then $S = \bigcap S_k$ is a surface with dimension $N - K$
- If $f|_S$ has a local minimum at $p \in S$ then $\nabla f(p) \perp T_p S$
- As before, we can prove that $\nabla g_k(p) \perp T_p S$
- Define the orthogonal space $\perp_p S = (T_p S)^\perp$
 - $T_p S$ is an $N - K$ dimensional plane
 - Therefore $\perp_p S$ is a subspace of dimension K
- Since $\nabla f(p), \nabla g_1(p), \dots, \nabla g_K(p)$ are $K + 1$ vectors in the K dimensional space $\perp_p S$ they are linearly dependent:

$$\nabla f(p) + \sum \lambda_k \nabla g_k(p) = 0$$



Tangent plane to surface S at P

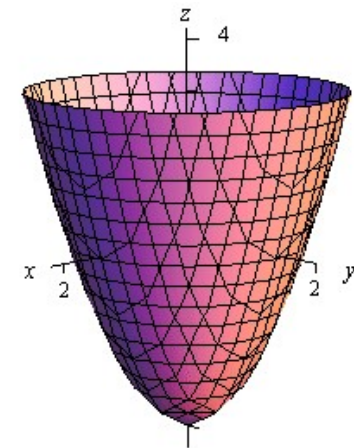
Taylor expansion and the Hessian matrix

- Let $\mathbf{x} = (x_1, \dots, x_n)$ and consider a twice differentiable function $f(\mathbf{x}): \mathbb{R}^n \rightarrow \mathbb{R}$
- The Taylor expansion of second order is

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{h} + \frac{1}{2} \mathbf{h}^T H[f](\mathbf{x}) \mathbf{h} + o(\|\mathbf{h}\|^2)$$

where:

$$H[f](\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix}$$



- The matrix $H[f](\mathbf{x})$ is called the Hessian (matrix)
- Since $f(\mathbf{x})$ is twice differentiable then $H[f](\mathbf{x})$ is a symmetric matrix
 - Therefore, $H[f](\mathbf{x})$ is **diagonalizable**.

Critical points and the Hessian matrix

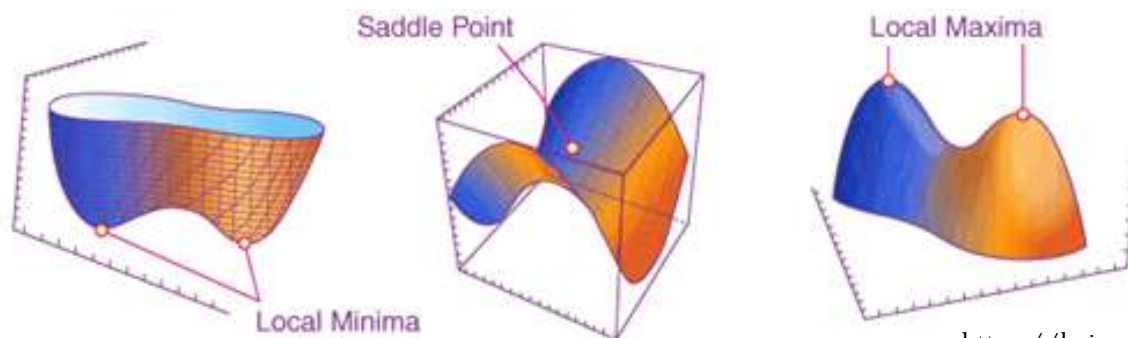
- Let $x_0 \in \mathbb{R}^n$ be a critical point of $f(x)$
- Then $\nabla f(x_0) = 0$ and the Taylor expansion becomes

$$f(x_0 + h) = f(x_0) + \frac{1}{2}h^T H[f](x_0)h + o(\|h\|^2)$$

- If $h \sim 0$ then

$$f(x_0 + h) \sim f(x_0) + \frac{1}{2}h^T H[f](x_0)h$$

- Based on the spectral properties of $H[f](x_0)$, we have the following statements:
 - If all of the eigenvalues of $H[f](x_0)$ are **positive** then x_0 is a local **minimum**
 - If all of the eigenvalues of $H[f](x_0)$ are **negative** then x_0 is a local **maximum**
 - If there are **positive and negative** eigenvalues of $H[f](x_0)$ then x_0 is a **saddle point**



<https://byjus.com/maths/saddle-points/>

Minimizing variance and optimising portfolios

- Let Q be the variance-covariance matrix:
 - It is symmetric and positive definite
- Consider the variance function $f(x) = x^T Q x$
 - It is quadratic and twice differentiable
- The Hessian matrix is Q
 - It is constant and all its eigenvalues are positive
- Therefore, any critical point of $f(x)$ is a local minimum
- This holds as well when we have constraints:
 - Minimum variance portfolio
 - Markowitz Portfolio
 - Etc...



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Morse theory

- Let us change coordinates to Z , the eigenvectors of $H[f](x_0)$:

$$f(z_0 + h) = f(z_0) + \frac{1}{2} h^T H[f](z_0) h + o(\|h\|^2)$$

- If the eigenvalues of $H[f](z_0)$ are $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ then

$$f(z_0 + h) = f(z_0) + \frac{1}{2} \sum_{j=1}^n \lambda_j h_j^2 + o(\|h\|^2)$$

- Suppose m eigenvalues are negative and $n - m$ are positive:

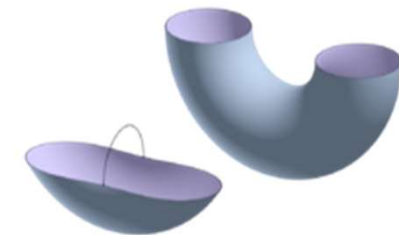
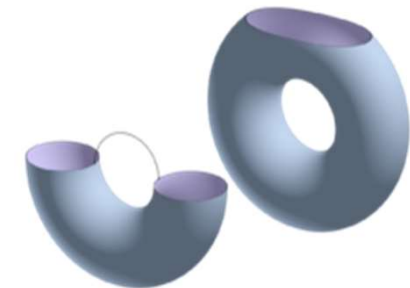
$$f(z_0 + h) = f(z_0) - \frac{1}{2} \sum_{j=1}^m |\lambda_j| h_j^2 + \frac{1}{2} \sum_{j=m+1}^n \lambda_j h_j^2 + o(\|h\|^2)$$

- We normalise the coordinates Z :

$$f(y_0 + h) = f(y_0) - \sum_{j=1}^m h_j^2 + \sum_{j=m+1}^n h_j^2 + o(\|h\|^2)$$

- Morse Lemma:** we can find **curvilinear coordinates** such that

$$f(y_0 + h) = f(y_0) - \sum_{j=1}^m h_j^2 + \sum_{j=m+1}^n h_j^2$$



Maximum Likelihood and the Hessian matrix 1

- Let $\mathbf{x} = (X_1, \dots, X_n)$ be i.i.d. Normal random variables i.e. $X_j \sim N(\mu, \sigma^2)$
- Their (joint) probability density function is

$$f(\mathbf{x}|\mu, \sigma^2) = \prod_{j=1}^n f_j(x_j|\mu, \sigma^2) = \left(\frac{1}{2n\sigma^2}\right)^{n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{j=1}^n (x_j - \mu)^2\right)$$

- Define $v = \sigma^2$ and the likelihood

$$\mathcal{L}(\mu, v) = \left(\frac{1}{2\pi v}\right)^{n/2} \exp\left(-\frac{1}{2v} \sum_{j=1}^n (x_j - \mu)^2\right)$$

- We want to **find** (μ_0, v_0) **that maximise** $\mathcal{L}(\mu, v)$.
- For easier computations, we will use the log-likelihood:

$$\ell(\mu, v) = \text{Log}(\mathcal{L}(\mu, v)) = -\frac{n}{2} \text{Log}(2\pi v) - \frac{1}{2v} \sum_{j=1}^n (x_j - \mu)^2$$

Maximum Likelihood and the Hessian matrix 2

- First, we compute the gradient of $\ell(\mu, v)$:

$$\frac{\partial \ell}{\partial \mu} = \frac{n}{v} \left(\frac{1}{n} \sum_{j=1}^n x_j - \mu \right)$$
$$\frac{\partial \ell}{\partial v} = \frac{n}{2v^2} \left(\frac{1}{n} \sum_{j=1}^n (x_j - \mu)^2 - v \right)$$



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- Second, we find the unique critical point (μ_0, v_0) of $\ell(\mu, v)$:

$$\frac{\partial \ell}{\partial \mu}(\mu_0, v_0) = 0 \leftrightarrow \mu_0 = \frac{1}{n} \sum_{j=1}^n x_j$$
$$\frac{\partial \ell}{\partial v}(\mu_0, v_0) = 0 \leftrightarrow v_0 = \frac{1}{n} \sum_{j=1}^n (x_j - \mu_0)^2$$

Maximum Likelihood and the Hessian matrix 3

- Third, we compute the Hessian of $\ell(\mu, v)$ at (μ_0, v_0) :

$$H[\ell](\mu_0, v_0) = \begin{bmatrix} \frac{\partial^2 \ell}{\partial \mu^2}(\mu_0, v_0) & \frac{\partial^2 \ell}{\partial \mu \partial v}(\mu_0, v_0) \\ \frac{\partial^2 \ell}{\partial v \partial \mu}(\mu_0, v_0) & \frac{\partial^2 \ell}{\partial v^2}(\mu_0, v_0) \end{bmatrix} = \begin{bmatrix} -\frac{n}{v_0} & 0 \\ 0 & -\frac{n}{2v_0^2} \end{bmatrix}$$

- Both eigenvalues of $H[\ell](\mu_0, v_0)$ are **negative**:
 - (μ_0, v_0) is a **global maximum** of $\ell(\mu, v)$
- The monotonicity of **Log** implies that (μ_0, v_0) is also a **global maximum** of $\mathcal{L}(\mu, v)$